

Survival MF Derivations - Yusha

Andrew Walther

January 29, 2026

1 Introduction

The following document presents a derivation of the objective function & algorithmic updates for supervised survival matrix factorization by Yusha Liu.

2 Objective Function

$$F(q_l, q_f, q_\beta, g_l, g_f, \beta, \tau) = E_{q_l, q_f, q_\beta} [\log p(Y, t, \delta | L, F, \beta, \tau)] + \bar{z}_k E_{q_{l_k}} \left[\log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)} \right] + \bar{z}_k E_{q_{f_k}} \left[\log \frac{g_{f_k}(f_k)}{q_{f_k}(f_k)} \right] + \bar{z}_k E_{q_{\beta_k}} \left[\log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)} \right] \quad (1)$$

$$\log P(Y, t, \delta | L, F, \beta, \tau) = \log P(Y | L, F, \sigma^2) \log P(t, \delta, L, \beta) \quad (2)$$

$$l(\eta) = \sum_i \delta_i \left[\eta_i - \log \left(\sum_{j \in R(t_i)} \exp(\eta_j) \right) \right] \quad (3)$$

$$= \left(\hat{\eta}^{(b-1)} \right) + \left(\eta - \hat{\eta}^{(b-1)} \right) \nabla l \left(\hat{\eta}^{(b-1)} \right) + \frac{1}{2} \left(\eta - \hat{\eta}^{(b-1)} \right)' \nabla^2 \left(\hat{\eta}^{(b-1)} \right) \left(\eta - \hat{\eta}^{(b-1)} \right) \quad (4)$$

$$= \left(\hat{\eta}^{(b-1)} \right) + \sum_i u_i^{(b-1)} \left(\eta_i - \hat{\eta}_i^{(b-1)} \right) + \frac{1}{2} \sum_i H_{ii}^{(b-1)} \left(\eta_i - \hat{\eta}_i^{(b-1)} \right)^2 \quad (5)$$

$$= - \sum_i \frac{1}{2} W_{ii} \left[\eta_i - \left(\hat{\eta}_i + \frac{u_i}{W_{ii}} \right)^2 \right] + u_i (\eta_i - \hat{\eta}_i) - \frac{1}{2} W_{ii} (\eta_i - \hat{\eta}_i)^2 \text{ (allowing } W_{ii} = -H_{ii}) \quad (6)$$

$$= - \frac{1}{2} W_{ii} [\eta_i^2 - 2\eta_i \hat{\eta}_i + \hat{\eta}_i^2 - 2W_{ii}^{-1} u_i (\eta_i - \hat{\eta}_i)] \quad (7)$$

$$= - \frac{1}{2} W_{ii} \left[\eta_i^2 - 2 \left(\hat{\eta}_i + \frac{u_i}{W_{ii}} \eta_i + \dots \right) \right] \quad (8)$$

$$= - \frac{1}{2} W_{ii} \left[\eta_i - \left(\hat{\eta}_i + \frac{u_i}{W_{ii}} \right) \right]^2 + \dots \quad (9)$$

$$F(q_l, q_f, q_\beta, g_l, g_f, \beta, \tau) = E_{q_l, q_f, q_\beta} [\log P(Y|L, F, \tau) \log P(t, \delta|L, \beta)] + \sum_k E_{q_{l_k}} \left[\log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)} \right] \quad (10)$$

$$+ \sum_k E_{q_{f_k}} \left[\log \frac{g_{f_k}(f_k)}{q_{f_k}(f_k)} \right] + E \left[\log \frac{g_\beta(\beta)}{q_\beta(\beta)} \right] \quad (11)$$

$$= E_{q_l, q_f, q_\beta} \left[\sum_i \sum_j -\frac{1}{2} \tau_j \left(y_{ij} - \sum_k l_{ik} f_{jk} \right)^2 + \sum_i -\frac{1}{2} W_{ii}^{(b-1)} \left(z_i^{(b-1)} - \sum_k l_{ik} \beta_k \right)^2 \right] \quad (12)$$

$$+ \sum_k \left(E_{q_{l_k}} \left[\log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)} \right] + E_{q_{f_k}} \left[\log \frac{g_{f_k}(f_k)}{q_{f_k}(f_k)} \right] \right) + E_{q_\beta} \left[\log \frac{g_\beta(\beta)}{q_\beta(\beta)} \right] + C \quad (13)$$

2A Update q_{l_k} (A.35 - A.40)

$$F(q_l, q_f, q_\beta, g_l, g_f, g_\beta, \tau) = E_{q_{l_k}} \left[-\frac{1}{2} \sum_i (A_{ik} l_{ik}^2 - 2B_{ik} l_{ik}) \right] + E_{q_{l_k}} \left[\log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)} \right] + C \quad (14)$$

- $A_{ik} = ???$
- $B_{ik} = ???$
- Simplified to EBNM problem
- Note/Hint: Drop the superscript $(b-1)$ for convenience.

$$W_{ii} \left(z_i - \sum_k l_{ik} \beta_k \right)^2 = W_{ii} (l_{ik}^2 \beta_k^2 - 2\tilde{z}_{ik} l_{ik} \beta_k) + C \quad (15)$$

where

$$\tilde{z}_{ik} = z_i - \sum_{k' \neq k} E_q [l_{ik'} \beta_{k'}] = z_i - \sum_{k' \neq k} \bar{l}_{ik'} \bar{\beta}_{k'} \quad (16)$$

Update 1/29/26

Objective Function for l_{ik} (drop superscript $(b-1)$)

$$F \approx E_q \left[\sum_{i,j} -\frac{1}{2} \tau_j (y_{ij} - \sum_k l_{ik} f_{jk})^2 + \sum_i -\frac{1}{2} W_{ii} (z_i - \sum_k l_{ik} \beta_k)^2 \right] \quad (17)$$

To update factor k , separate the sum $\sum_k$ into the current factor k and all other factors $k' \neq k$. Let the residuals for the other factors be:

- **MF Residual:** $R_{ij}^{-k} = y_{ij} - \sum_{k' \neq k} l_{ik'} f_{jk'}$
- **Survival Residual:** $\tilde{z}_i^{-k} = z_i - \sum_{k' \neq k} l_{ik'} \beta_{k'}$

Sub these into the objective function F :

$$F(l_{ik}) \propto E_q \left[\sum_j -\frac{1}{2} \tau_j (R_{ij}^{-k} - l_{ik} f_{jk})^2 - \frac{1}{2} W_{ii} (\tilde{z}_i^{-k} - l_{ik} \beta_k)^2 \right] \quad (18)$$

Now expand the quadratic terms, retaining only the terms involving l_{ik} :

$$(R_{ij}^{-k} - l_{ik} f_{jk})^2 = (R_{ij}^{-k})^2 - 2R_{ij}^{-k} l_{ik} f_{jk} + l_{ik}^2 f_{jk}^2 \quad (19)$$

$$(\tilde{z}_i^{-k} - l_{ik} \beta_k)^2 = (\tilde{z}_i^{-k})^2 - 2\tilde{z}_i^{-k} l_{ik} \beta_k + l_{ik}^2 \beta_k^2 \quad (20)$$

Group the terms and apply expectations then substitute into F :

$$F(q_{l_k}) \propto \sum_j -\frac{1}{2}\tau_j (l_{ik}^2 E[f_{jk}^2] - 2l_{ik} E[f_{jk}] E[R_{ij}^{-k}]) - \frac{1}{2}W_{ii} (l_{ik}^2 E[\beta_k^2] - 2l_{ik} E[\beta_k] \bar{z}_i^{-k}) \quad (21)$$

Then group the terms according to their power on l_{ik} :

$$F(q_{l_k}) \propto -\frac{1}{2}l_{ik}^2 \underbrace{\left(\sum_j \tau_j \bar{f}_{jk}^2 + W_{ii} \bar{\beta}_k^2 \right)}_{A_{ik}} + l_{ik} \underbrace{\left(\sum_j \tau_j \bar{f}_{jk} E[R_{ij}^{-k}] + W_{ii} \bar{\beta}_k \bar{z}_i^{-k} \right)}_{B_{ik}} \quad (22)$$

This yields the following results for A_{ik} and B_{ik} :

$$\begin{aligned} A_{ik} &= \sum_j \tau_j \bar{f}_{jk}^2 + W_{ii} \bar{\beta}_k^2 \\ B_{ik} &= \sum_j \tau_j \bar{f}_{jk} (y_{ij} - \sum_{k' \neq k} \bar{l}_{ik'} \bar{f}_{jk'}) + W_{ii} \bar{\beta}_k (z_i - \sum_{k' \neq k} \bar{l}_{ik'} \bar{\beta}_{k'}) \end{aligned}$$

2B Update q_{f_k}

$$F(q_l, q_f, q_\beta, q_l, g_f, g_\beta, \tau) = E_{q_{f_k}} \left[-\frac{1}{2} \sum_j \left(\tilde{A}_{jk} f_{jk}^2 - 2\tilde{B}_{jk} f_{jk} \right) \right] + E_{q_{f_k}} \left[\log \frac{g_{f_k}(f_k)}{q_{f_k}(f_k)} \right] + C \quad (23)$$

- $\tilde{A}_{jk} = ???$
- $\tilde{B}_{jk} = ???$

Update 1/29/26

Isolate the objective function for f_{ik} . These factors only appear in the MF term, not in the survival term (ignore survival term):

$$F(f_{jk}) \propto E_q \left[\sum_i -\frac{1}{2} \tau_j (y_{ij} - \sum_k l_{ik} f_{jk})^2 \right] \quad (24)$$

Then include the residuals

$$F(f_{jk}) \propto E_q \left[\sum_i -\frac{1}{2} \tau_j (R_{ij}^{-k} - l_{ik} f_{jk})^2 \right] \quad (25)$$

and expand the square for terms with f_{ik} :

$$\propto -\frac{1}{2} \tau_j (l_{ik}^2 f_{jk}^2 - 2l_{ik} f_{jk} R_{ij}^{-k}) \quad (26)$$

Apply expectations

$$F(q_{f_k}) \propto -\frac{1}{2} f_{jk}^2 \left(\tau_j \sum_i E[l_{ik}^2] \right) + f_{jk} \left(\tau_j \sum_i E[l_{ik}] E[R_{ij}^{-k}] \right) \quad (27)$$

Then identify terms $\tilde{A}_{jk}$ and $\tilde{B}_{jk}$ matching the form $-\frac{1}{2}(\tilde{A}_{jk} f_{jk}^2 - 2\tilde{B}_{jk} f_{jk})$:

$$\tilde{A}_{jk} = \tau_j \sum_i \bar{l}_{ik}^2 \quad (28)$$

$$\tilde{B}_{jk} = \tau_j \sum_i \bar{l}_{ik} (y_{ij} - \sum_{k' \neq k} \bar{l}_{ik'} \bar{f}_{jk'}) \quad (29)$$

2C Update q_β

Let (Expectation - Variance relationship)

- $\bar{l}_{ik} = E_{q_l} [l_{ik}]$
- $\bar{l}_{ik}^2 = E_{q_l} [L_{ik}^2]$
- $\bar{l}_{ik}^2 = (\bar{l}_{ik})^2 + Var_{q_l} [l_{ik}]$

$$F(q_l, q_f, q_\beta, g_l, g_f, g_\beta, \tau) \quad (30)$$

$$= E_{q_l, q_\beta} \left[-\frac{1}{2} \sum_i W_{ii}^{(b-1)} \left(\sum_k l_{ik}^2 \beta_k^2 + 2 \sum_{k \neq k'} l_{ik} \beta_k l_{ik'} \beta_{k'} - 2z_i^{(b-1)} \sum_k l_{ik} \beta_k \right) \right] \quad (31)$$

$$+ E_{q_\beta} \left[\log \frac{g_\beta(\beta)}{q_\beta(\beta)} \right] + Constant \quad (32)$$

$$= E_{q_\beta} \left[-\frac{1}{2} \sum_i W_{ii}^{(b-1)} \left(\sum_k \bar{l}_{ik}^2 \beta_k^2 + 2 \sum_{k \neq k'} \bar{l}_{ik} \bar{l}_{ik'} \beta_k \beta_{k'} - 2z_i^{(b-1)} \sum_k \bar{l}_{ik} \beta_k \right) \right] \quad (33)$$

$$+ E_{q_\beta} \left[\log \frac{g_{\beta\beta}(\beta)}{q_{\beta\beta}(\beta)} \right] + Constant \quad (34)$$

$$= E_{q_\beta} \left[-\frac{1}{2} \sum_i W_{ii}^{(b-1)} \left(\sum_k (\bar{l}_{ik})^2 \beta_k^2 + 2 \sum_{k \neq k'} \bar{l}_{ik} \bar{l}_{ik'} \beta_k \beta_{k'} - 2z_i^{(b-1)} \sum_k \bar{l}_{ik} \beta_k \right) - \frac{1}{2} \sum_i W_{ii}^{(b-1)} \left(\sum_k Var_{q_l}(l_{ik}) \beta_k^2 \right) \right] \quad (35)$$

$$+ E_{q_\beta} \left[\log \left(\pi_k \frac{1}{\sqrt{2\pi\sigma}} \exp \left(-\frac{1}{2\sigma^2} \beta_k^2 \right) \right) \right] - E_{q_\beta} [\log q(\beta)] \quad (36)$$

$$= E_{q_\beta} \left[-\frac{1}{2} \sum_i W_{ii}^{(b-1)} \left(z_i^{(b-1)} - \sum_k \bar{l}_{ik} \beta_k \right)^2 \right] \quad (37)$$

$$+ E_{q_\beta} \left[-\frac{1}{2} \sum_k \left(\frac{1}{\sigma^2} + \sum_i W_{ii}^{(b-1)} Var_{q_l}(l_{ik}) \right) \beta_k^2 \right] - E_{q_\beta} [\log q(\beta)] \quad (38)$$

This is the same objective function as

$$z_i^{(b-1)} = \sum_k \bar{l}_{ik} \beta_k + \varepsilon_i, \quad \varepsilon_i \sim N \left(0, \frac{1}{W_{ii}^{(b-1)}} \right) \quad (39)$$

with the prior

$$\beta_k \sim N \left(0, \frac{1}{\frac{1}{\sigma^2} + \sum_i W_{ii}^{(b-1)} Var_{q_l}(l_{ik})} \right) \quad (40)$$

Recall that $\eta_i = \sum_k l_{ik} \beta_k$ so after iteration (b), update $\hat{\eta}_i^{(b)} = E_{q_l, q_\beta} [\sum_k l_{ik} \beta_k] = \sum_k \bar{l}_{ik} \bar{\beta}_k$

Update 1/29/26

Identify the objective function for β - including both the survival term and the prior distribution term

$$F(\beta) \propto E_q \left[\sum_i -\frac{1}{2} W_{ii} (z_i - \sum_k l_{ik} \beta_k)^2 \right] + \sum_k \left(-\frac{1}{2\sigma^2} \beta_k^2 \right) \quad (41)$$

Then isolate β_k for all β_k where $k \neq k'$ and let the survival residual be $\tilde{z}_i^{-k} = z_i - \sum_{k \neq k'} l_{ik'} \beta_{k'}$:

$$F(\beta_k) \propto E_q \left[\sum_i -\frac{1}{2} W_{ii} (\tilde{z}_i^{-k} - l_{ik} \beta_k)^2 \right] - \frac{1}{2\sigma^2} \beta_k^2 \quad (42)$$

Expand and group quadratic terms:

$$(\tilde{z}_i^{-k} - l_{ik}\beta_k)^2 \rightarrow l_{ik}^2\beta_k^2 - 2l_{ik}\beta_k\tilde{z}_i^{-k} + C \quad (43)$$

Then substitute back into F :

$$F(\beta_k) \propto \sum_i -\frac{1}{2}W_{ii} (E[l_{ik}^2]\beta_k^2 - 2E[l_{ik}]\beta_k\tilde{z}_i^{-k}) - \frac{1}{2\sigma^2}\beta_k^2 \quad (44)$$

And group by β_k powers:

$$F(\beta_k) = -\frac{1}{2}\beta_k^2 \underbrace{\left(\frac{1}{\sigma^2} + \sum_i W_{ii}\bar{l}_{ik}^2\right)}_{\text{Posterior Precision}} + \beta_k \underbrace{\left(\sum_i W_{ii}\bar{l}_{ik}\tilde{z}_i^{-k}\right)}_{\text{Posterior Precision} \times \text{Mean}} \quad (45)$$

Then the updates for q_{β_k} are a normal distribution with the following parameters:

- **Variance** ($\sigma_{\beta_k}^2$):

$$\frac{1}{\sigma_{\beta_k}^2} = \frac{1}{\sigma^2} + \sum_i W_{ii}\bar{l}_{ik}^2$$

- **Mean** (μ_{β_k}):

$$\mu_{\beta_k} = \sigma_{\beta_k}^2 \left(\sum_i W_{ii}\bar{l}_{ik}\tilde{z}_i^{-k} \right)$$

2D Regression Model

Consider the linear regression model

$$z_i = X_i'\beta + \varepsilon_i, \quad \varepsilon_i \sim N(0, W_{ii}^{-1}) \quad (46)$$

where the prior is

$$\beta_k \stackrel{\text{iid}}{\sim} N(0, \sigma^2) \Rightarrow \beta | \sigma^2 \sim N(0, \sigma^2 I_{k \times k}) \quad (47)$$

and posterior

$$q(\beta) \sim N(\text{Mean}, \text{Variance}). \quad (48)$$

In an EB framework

$$F(q_\beta, g_\beta) \quad (49)$$

$$= E_q[\log P(z|\beta)] + E_q\left[\log \frac{g_\beta(\beta)}{q_\beta(\beta)}\right] \quad (50)$$

$$= E_q\left[\log\left(\pi_i \frac{1}{\sqrt{2\pi}} W_{ii}^{\frac{1}{2}} \exp\left[-\frac{1}{2}W_{ii}(z_i - x_i'\beta)^2\right]\right)\right] + E_q\left[\log\left(\pi_k \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2\sigma^2}\beta_k^2\right]\right)\right] \quad (51)$$

$$- E_q[\log q(\beta)] \quad (52)$$

$$= E_q\left[-\frac{1}{2}\sum_i W_{ii}(z_i - x_i'\beta)^2 - \frac{1}{2\sigma^2}\sum_k \beta_k^2 - \log q_\beta(\beta)\right] \quad (53)$$